

Course Type	Course Code	Course Name	Credits
HMC	HMCAL703	DEEP LEARNING	04

Examination Scheme					
Distribution of Marks			Exam Duration (Hrs.)		Total Marks
In-semester Assessment		End Semester Examination (ESE)			
Continuous Assessment	Mid-Semester Exam (MSE)		MSE	ESE	
20	30	50	1.5	2	100

Pre-requisite:

1. HMCAL501- Knowledge Engineering
2. HMCAL602- Foundation of Machine Learning

Program Outcomes addressed:

1. PO1: Engineering knowledge
2. PO2: Problem Analysis
3. PO3: Design/Development of Solutions
4. PO5: Engineering Tool Usage
5. PO6: The Engineer & The World
6. PO11: Life-long learning

Course Objectives:

1. To acquaint with the fundamentals of Neural Network.
2. To facilitate in depth understanding of training Deep Neural Networks.
3. To acquire knowledge of Convolution Neural Networks, Auto encoders and Recurrent Neural Networks.
4. To encourage in exploring emerging trends in Deep Learning to develop intelligent applications.

Module	Detailed Contents	Hrs.
	Course Introduction This course provides students with a comprehensive understanding of deep learning, focusing on both foundational concepts and advanced architectures. Students will gain hands-on and conceptual proficiency in key deep learning paradigms, including Convolutional Neural Networks (CNNs) for supervised learning, Auto encoders for unsupervised representation learning, and Recurrent Neural Networks (RNNs) for sequence modelling. The course also explores recent trends such as Generative Adversarial Networks (GANs), preparing students to apply deep learning methods to real-world, research-oriented, and industry-driven applications in an evolving AI landscape.	01
01.	Fundamentals of Neural Network <i>Learning Objective:</i> To explore the composition of biological neuron underlying the structure of deep neural network, key terminologies and historical evolution of deep learning.	4-6

	Contents: • Fundamentals of Neural Network: Biological neuron • Mc-Culloch Pitts Neuron, Perceptron, Multilayer Perceptron • Perceptron Learning, Delta learning • Linearly separable, linearly non-separable classes • Deep Networks: Fundamentals, Brief History • Basic Terminologies of Deep Learning • Classes of Deep Learning	
	Self-Learning Topics: Transfer Learning Learning Outcomes: A learner will be able to LO 1.1: Apply linear-algebraic and discrete-mathematical techniques to model and interpret the mathematical representation of biological neurons and the McCulloch–Pitts neuron. (P.I.-1.1.1) LO 1.2: Apply principles of computer engineering to determine linear separability of given dataset using perceptron-based model. (P.I - 1.4.1) LO 1.3: Apply foundational and specialized engineering concepts to classify core terminologies and major categories of Deep Learning models.(P.I.-1.3.1)	
02.	Training, Optimization and Regularization of Deep Neural Network Learning Objective: To construe the training of deep networks, analyze and apply appropriate learning parameters and optimization techniques, and evaluate and implement effective regularization strategies for improved model performance. Contents: • Training Feedforward DNN: Multi Layered Feed Forward Neural Network, Learning Factors • Activation functions: Tanh, Logistic, Linear, Softmax, ReLU, Leaky ReLU • Loss functions: Squared Error loss, Cross Entropy • Choosing output function and loss function • Optimization: Learning with backpropagation • Learning Parameters: Gradient Descent (GD), Stochastic and Mini Batch GD, Momentum Based GD, Nesterov Accelerated GD, AdaGrad, Adam, RMSProp. • Regularization: Overview of Overfitting, Types of biases, Bias Variance Tradeoff. • Regularization Methods: L1, L2 regularization, Parameter sharing, Dropout, Weight Decay, Batch normalization, Early stopping, Data Augmentation, adding noise to input and output.	10-12

	Self-Learning Topics: <i>Advanced Regularization Techniques in Modern Deep Learning Architectures.</i> <i>Hyperparameter Tuning Strategies for Efficient Deep Learning Model Optimization.</i> Learning Outcomes: A learner will be able to LO 2.1: Apply linear-algebraic and discrete-mathematical techniques to mathematically represent Multi Layered Feed Forward Neural Network. (P.I.-1.1.1) LO 2.2: Apply engineering knowledge to formulate computational modules with suitable activation and loss functions for a given problem context. (P.I.-1.3.1) LO 2.3: Identify optimization algorithms of a computer based system and tune learning parameters by applying mathematical foundations of gradient-based methods to Enhance convergence and model performance. (P.I.- 2.1.2) LO 2.4: Compare and contrast alternative methods to select appropriate regularization technique to reduce over-fitting and improve model generalization using engineering fundamentals. (P.I.- 2.2.4)	
03.	Convolutional Neural Networks (CNN): Supervised Learning Learning Objective: <i>To interpret supervised learning concepts in CNNs, apprehend convolution operations and architectural components, and interpret foundational models for effective feature extraction and classification.</i>	8-10
	Contents: • Supervised Learning: Convolution operation, Padding, Stride, Relation between input, output and filter size • CNN architecture: Convolution layer, Pooling Layer, Weight Sharing in CNN • Fully Connected NN vs CNN • Variants of basic Convolution function • LeNET: Architecture • AlexNET: Architecture Self-Learning Topics: <i>Evolution of CNN Architectures: From VGG to ResNet</i> <i>Advanced CNN Architectures</i> Learning Outcomes: LO 3.1: Apply fundamental knowledge of mathematics to compute convolution, padding and stride operations in CNNs. (P.I.- 1.1.1) LO 3.2: Apply theory and principles of supervised learning to determine the relationship between input, output, and filter size in CNNs. (P.I.- 1.3.1) LO3.3: Design a variety of potential CNN architecture design solutions suited for specified image data to meet functional requirements. (P.I.- 3.2.2) LO 3.4: Explore classical architectural designs for basic image-classification problems. (P.I.-3.2.1)	
04.	Autoencoders: Unsupervised Learning Learning Objective: <i>To construe the concepts of unsupervised learning and auto-encoder models for image processing tasks such as image compression and feature representation.</i>	8-10

	Contents: • Unsupervised Learning: Introduction • Linear Autoencoder, Undercomplete Autoencoder, Overcomplete Autoencoders, Denoising Autoencoders, Sparse Autoencoders, Contractive Autoencoders • Autoencoders bottleneck and its impact. • Application of Auto-encoders: Image Compression, Image Denoising, Dimensionality Reduction.	
	Self-Learning Topics: Variational Autoencoders (VAEs) Learning Outcomes: A learner will be able to LO 4.1: Apply fundamental knowledge of neural networks to interpret the working principles of autoencoder. (P.I.-1.3.1) LO 4.2: Apply foundational and specialized engineering concepts to categorize major autoencoder models. (P.I.-1.4.1) LO 4.3: Explore and design suitable autoencoder variants for image compression or denoising pipelines by evaluating multiple architectural alternatives. (P.I.-3.2.1) LO 4.4: Able to verify the functionalities influencing representation capacity and reconstruction quality for different datasets and validate design of auto-encoders. (P.I.-3.4.3)	
05.	Recurrent Neural Networks (RNN) Learning Objective: To construe the concepts of sequence learning that enhances usability of application.	10-12
	Contents: • Sequence Learning Problem, Unfolding Computational graphs • Recurrent Neural Network, Bidirectional RNN • Backpropagation Through Time (BTT), Vanishing and Exploding Gradients, Truncated BTT • Long Short-Term Memory: Selective Read, Selective write, Selective Forget, Gated Recurrent Unit. Self-Learning Topics: Attention Mechanisms in RNNs RNNs in Reinforcement Learning Learning Outcomes: A learner will be able to LO 5.1: Apply fundamentals of engineering to solve sequencing learning problems. (P.I.-1.3.1) LO 5.2: Apply computer engineering principles to construe with working principles of Recurrent Neural Networks. (P.I. - 1.4.1) LO 5.3: Analyze and interpret the effect of vanishing and exploding gradient problems on learning performance in sequence modeling tasks. (P.I.- 2.4.2) LO 5.4: Analyze sequence-learning problems to identify and conform suitable RNN variants and assess the strengths and limitations of relevant tools for their implementation. (P.I.- 2.1.2, 5.1.2, 5.2.1) LO 5.5: Interpret legislation and regulations relevant for the protection of the public and understand the relationship between the technical, socio-economic and environmental dimensions of sustainability while deploying RNN-based systems. (P.I.-6.2.1, 6.3.2)	

06.	Emerging Trends <i>Learning Objective:</i> Utilize emerging trends that enhance usability of generative models	6-8
	Contents: - Generative Adversarial Network (GAN) Architecture - Image Generation and DeepFake - Diffusion Models for Image Generation	
	Self-Learning Topics: Advanced GAN Variants: StyleGAN, CycleGAN, and BigGAN. Responsible AI Practices for Generative Models Learning Outcomes: A learner will be able to LO 6.1: Apply engineering fundamentals to interpret GAN architecture and training dynamics. (P.I.-1.3.1) LO 6.2: Apply theory and principles of computer science and engineering to generate synthetic images with GAN architecture. (P.I.-1.4.1) LO 6.3: Recognize ethical, legal and societal implications and understand the environmental dimensions of sustainability for image generation using Deep Fake technique. (P.I.-6.2.1, 6.3.2) LO 6.4: Demonstrate continuous professional learning by tracking developments in generative models, including diffusion-based image generation techniques, and analyzing sourced technical information for feasibility and sustainability to propose diverse applications. (P.I.-11.2.2, 11.3.2)	
	Course Conclusion: Upon completion of the Deep Learning course, students will acquire the proficiency to design, train, and apply advanced neural network models to solve real-world, research-oriented, and industry-focused problems.	01
Total		60

Performance Indicators:

P.I No	P.I. Statement
1.1.1	Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems.
1.3.1	Apply engineering fundamentals
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a Problem.
2.1.3	Identify mathematical algorithmic knowledge that applies to a given problem
2.2.4	Compare and contrast alternative solution/methods to select the best methods
2.4.2	Analyze and interpret the results using contemporary tools.
3.2.1	Able to explore design alternatives.
3.2.2	Able to produce a variety of potential design solutions suited to meet functional requirement
3.4.3	Able to verify the functionalities and validate the design
5.1.2	Adapt the tools and techniques to solve engineering problems.
6.2.1	Interpret legislation, regulations, codes, and standards relevant to your discipline and explain its contribution to the protection of the public.
6.3.2	Understand the relationship between the technical, socio-economic and environmental dimensions of sustainability.

- 11.2.2 Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in your field
- 11.3.2 Analyze sourced technical and popular information for feasibility, viability, sustainability etc.

Course Outcomes: A learner will be able to-

1. Apply engineering and mathematical foundations to interpret mathematical representation of biological neurons, differentiate linearly separable and linearly non-separable classes using perceptron-based methods, and categorize core terminologies and principal categories of deep learning models.(LO 1.1, LO 1.2, LO 1.3)
2. Formulate modules of a system with appropriate activation and loss functions for a given problem context, identify optimization algorithms to enhance model performance, and analyze alternative regularization techniques for more generalized model.(LO 2.1, LO 2.2, LO 2.3, LO 2.4)
3. Design variety of potential CNN architectures suited for specified image data and explore classical architectural designs- LeNet and AlexNet for image-classification problems.(LO 3.1, LO 3.2, LO 3.3, LO 3.4)
4. Design components of auto-encoder variants for image compression or denoising pipelines and dimensionality reduction across different datasets.(LO 4.1, LO 4.2, LO 4.3, LO 4.4)
5. Apply engineering principles to solve sequence learning problems by adapting appropriate Recurrent Neural Network (RNN) variants and tools, while considering ethical, legal, and sustainability impacts on society and the environment.(LO 5.1, LO 5.2, LO 5.3, LO 5.4, LO 5.5)
6. Apply principles of computer science and engineering to evaluate Generative Adversarial Network (GAN) architectures for synthetic image generation, while continuously tracking emerging trends with consideration of ethical, legal, and sustainability impacts on society and the environment.(LO 6.1, LO 6.2, LO 6.3, LO 6.4)

CO-PO Mapping Table with Correlation Level

CO ID	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
HMCAL703.1	3	--	--	--	--	--	--	--	--	--	--
HMCAL703.2	3	3	--	--	--	--	--	--	--	--	--
HMCAL703.3	3	--	3	--	--	--	--	--	--	--	--
HMCAL703.4	3	--	3	--	--	--	--	--	--	--	--
HMCAL703.5	3	3	--	--	3	3	--	--	--	--	--
HMCAL703.6	3	--	--	--	--	3	--	--	--	--	3
Average	3	3	3	--	3	3	--	--	--	--	3

Text Books:

1. Ian Goodfellow, Yoshua Bengio, Aaron Courville. —Deep Learning, MIT Press Ltd, 2016
2. Li Deng and Dong Yu, —Deep Learning Methods and Applications||, Publishers Inc.
3. Satish Kumar "Neural Networks A Classroom Approach" Tata McGraw-Hill.
4. JM Zurada —Introduction to Artificial Neural Systems||, Jaico Publishing House
5. M. J. Kochenderfer, Tim A. Wheeler. —Algorithms for Optimization||, MIT Press.

Reference Books:

1. Fundamentals of deep learning: Designing next-generation machine intelligence algorithms, Buduma, N. and Locascio, N.,2017, O'Reilly Media, Inc.
2. Deep learning with Python ,François Chollet,(Vol. 361). 2018 New York,Manning.
3. Deep Learning Cookbook, Douwe Osinga,O'REILLY, SPD Publishers, Delhi.
4. Neural Network- A Comprehensive Foundation,Simon Haykin, Prentice Hall International, Inc
5. Principles of soft computing,S.N.Sivanandam and S.N.Deepa, Wiley India

Other Resources:

1. <https://nptel.ac.https://deeplearning.cs.cmu.edu/S21/index.html>
2. <http://www.cse.iitm.ac.in/~miteshk/CS6910.html>
3. <https://nptel.ac.in/courses/106/106/106106184/>
4. <https://www.deeplearningbook.org>

IN-SEMESTER ASSESSMENT (50 MARKS)**1. Continuous Assessment (20 Marks)**

1. Assignment on live problems/case studies: 10 Marks
(Assignment on live problems/ case studies, wherein problems are given prior. Students are expected to research and collect required resources. They can use the resources and solve the problem on assigned date and time in Institute premises in presence of faculty member.)
2. Article reading and summarization: 05 Marks
3. Regularity and active participation: 05 Marks

2. Mid Semester Exam (30 Marks)

Mid semester examination will be based on 40% to 50% syllabus.

END SEMESTER EXAMINATION (50 MARKS)

End Semester Examination will be based on syllabus coverage up to the Mid Semester Examination(MSE) carrying 20%-30% weightage, and the syllabus covered from MSE to ESE carrying 70%-80% weightage.